



PORTFOLIO · GROUNDED AI

Demo — DCA Investment Planner

Senior Data Engineer · Etops

Surapat Ek-In · CoreSat demo

Goal, data & stack



Goal

Provide a DCA investment plan and projected future growth.



Data

yfinance · SEC EDGAR · PDF documents



Stack

BACKEND

Python · FastAPI

DATABASE

PostgreSQL · pgvector

LLM

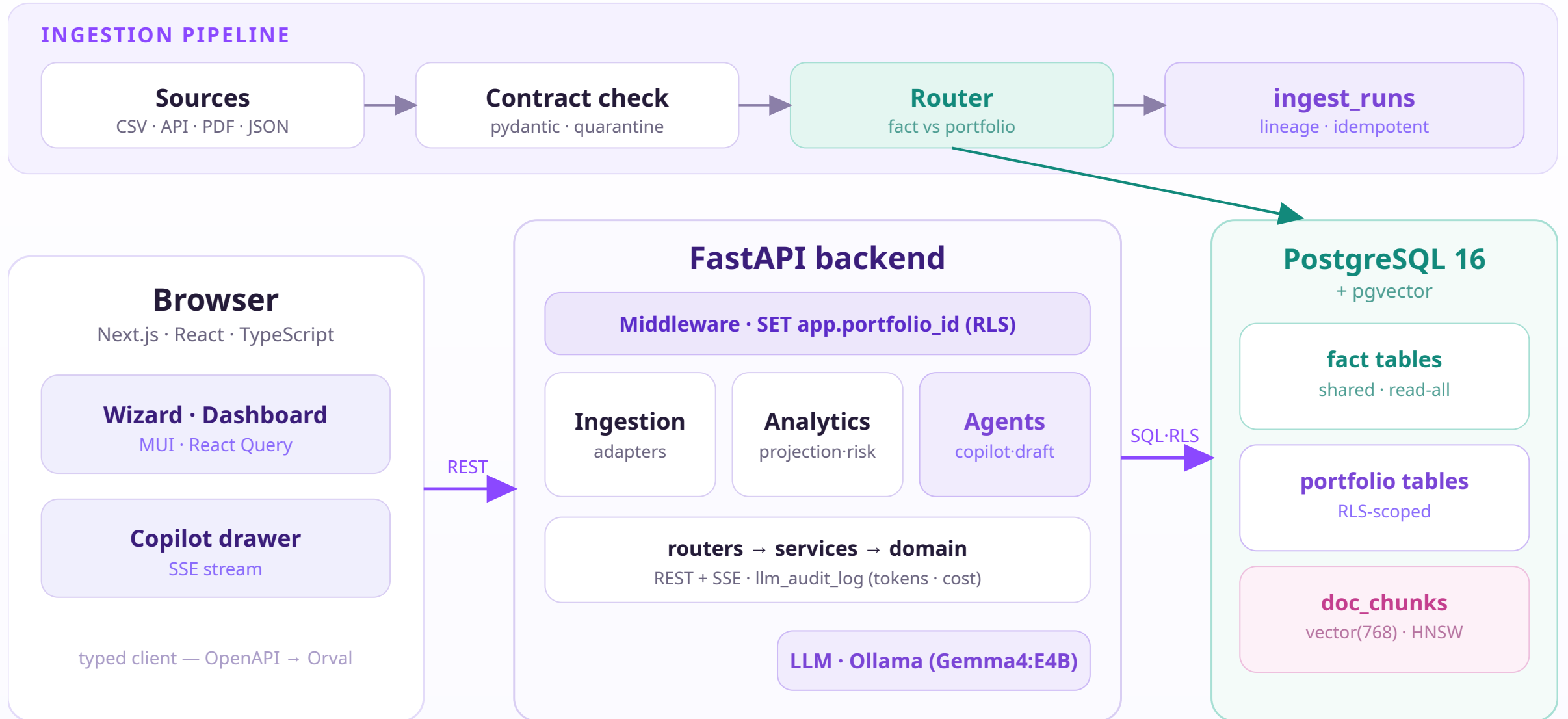
Gemma4:E4B

FRONTEND

React · TypeScript · MUI

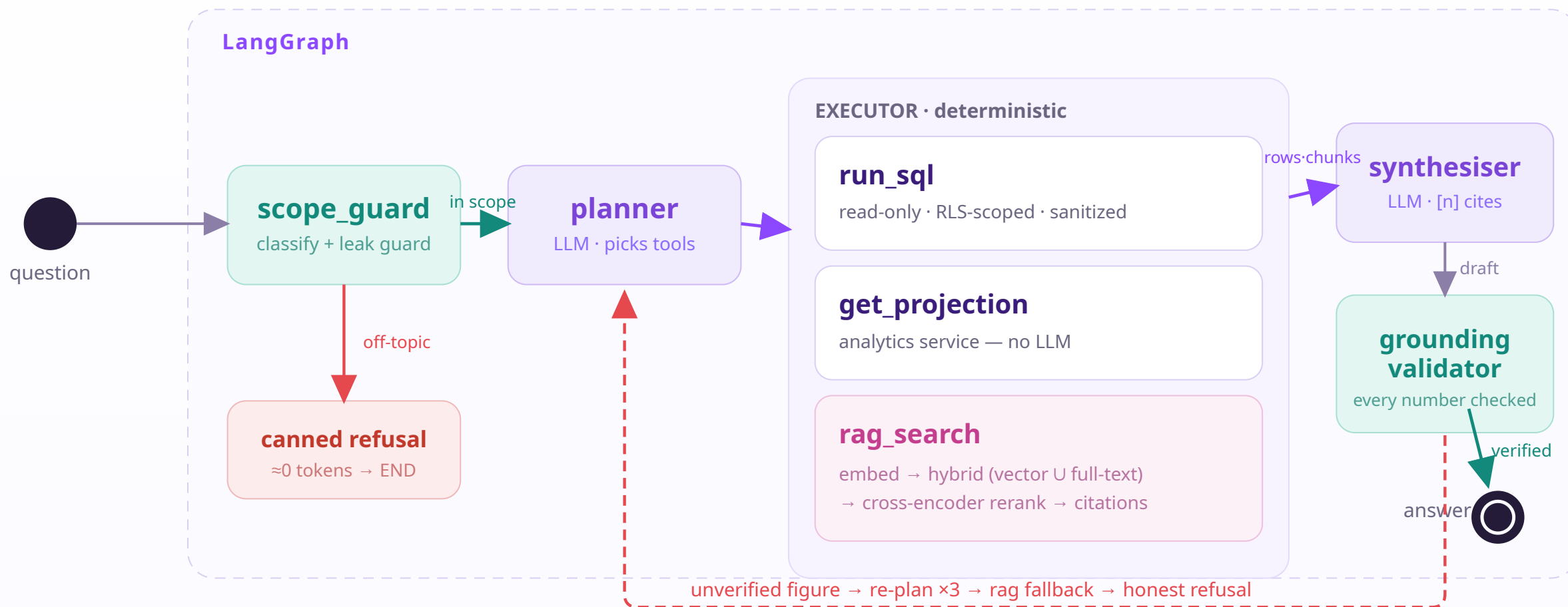
System architecture

Next.js → FastAPI → one PostgreSQL · RLS isolation



Copilot agent

the model writes queries, never numbers



LLM in exactly 3 nodes · retrieval and every number are deterministic

a prompt-injected query still cannot leave the portfolio

Status & my first 90 days

Demo

- LLM usage in an app
- Python · FastAPI backend
- React · TypeScript frontend

To improve

- Evaluation
grounding accuracy
- Speed
latency · cost

First 3 months at Etops

- Understand the data team's stack
- Find critical parts to improve
- Take ownership of them